

Chapter 106 — Evaluation and Management of Males With Urinary Incontinence

URINARY INCONTINENCE

Definition and Determinants

- **Urinary incontinence (UI)** — International Continence Society (ICS): **the complaint of any involuntary loss of urine**. It is a **storage** lower urinary tract symptom (LUTS); LUTS are storage, voiding or postmicturition (Table 106.1).
- **Male continence** depends on the external urethral sphincter (EUS), internal urethral sphincter (IUS), supporting structures and innervation — **the EUS is the most important determinant**.
- **Risk factors: age (major and independent)**, LUTS, infection, functional and cognitive impairment, neurological disease, obesity, diabetes, prostatectomy, drugs, smoking, alcohol.
- **DIAPPERS** — commonest causes, especially in the elderly: **Delirium, Infection, Atrophic urethritis, Pharmacological agents, Psychiatric illness, Endocrine disorders, Reduced mobility, Stool impaction**.
- **Male UI is more often multifactorial** than female UI → establish aetiology before treating; it carries social, economic and hygienic burden and impairs quality of life (QoL).
- **Describe UI by type, frequency, severity, precipitants, social, hygiene and QoL impact, containment, and whether help is wanted. Distinguish from sweating and urethral discharge.**

Epidemiology

- **Less prevalent than in females — male:female ratio about 1:2** — and less studied; male incontinence is described in Ancient Egyptian manuscripts about 4000 years ago.
- **Reported prevalence 1–40%**, the spread reflecting definition, type, population, age and survey method.
- **Urgency urinary incontinence (UUI) predominates** in men, followed by stress urinary incontinence (SUI), then mixed urinary incontinence (MUI).
- **EpiLUTS** reported a far higher male prevalence (46%) because it counted postmicturition incontinence, nocturnal enuresis and leakage without definable cause — **definition drives the figure**.
- **Rises with age**: about 3–5% in young and middle-aged men versus 11–34% in older men — roughly one in ten at 60–64 years, approaching one in three over 85 years.
- **Race and ethnicity**: the National Health and Nutrition Examination Survey (NHANES) found no effect; other studies found one — **highest in non-Hispanic Black men, lowest in Chinese men**, with UUI the dominant type.

KEY POINT: RISK FACTORS FOR URINARY INCONTINENCE IN MALES

- **Age** (major, independent), LUTS, urological infection, functional and cognitive impairment, neurological disease, diabetes, drugs, smoking, alcohol, and **surgery (prostatectomy)**.

Concept — *Male incontinence is age-driven, multifactorial and dominated by urgency; stress incontinence in men is largely an iatrogenic sphincter injury, which is why evaluation hunts for the cause rather than assuming a single mechanism.*

TYPES OF URINARY INCONTINENCE

KEY POINTS: TYPES OF URINARY INCONTINENCE IN MALES (ICS DEFINITIONS)

- **SUI** — involuntary loss on effort or physical exertion, or on sneezing, laughing or coughing, **not caused by a bladder contraction**.
- **UII** — involuntary loss **associated with urgency**.
- **MUI** — involuntary loss associated with urgency **and** with effort.
- **Nocturnal enuresis** — intermittent incontinence **during sleep**.
- **Continuous UI** — continuous involuntary loss **without normal voiding**.
- **UI associated with chronic retention** — from **failure of the bladder to empty**.

TYPES OF URINARY INCONTINENCE — MECHANISM AND MALE CONTEXT

Type	Mechanism	Male context and points of note
SUI	Intra-abdominal pressure exceeds intraurethral pressure , usually with sphincter deficiency	Usually only after prostatectomy (EUS damage); commoner in females. "Effort-" or "activity-related" incontinence is preferred in some languages to avoid confusion with psychological stress
UII	Detrusor overactivity (DO) raises intravesical pressure above outlet resistance; about 90% of men with UII have DO on urodynamics	= wet overactive bladder (OAB) . Renamed from "urge" to "urgency" incontinence because it must be preceded by urgency
MUI	Both DO and sphincter deficiency	Uncommon in men ; may follow prostatectomy
Nocturnal enuresis	In an adult man, may signal high-pressure chronic retention with upper-tract dilatation and risk of renal failure	About 10% of 7-year-olds; persists into adulthood in 2–3% of children; may also arise de novo later in life
Continuous UI	Fistula (e.g., rectoprostatic), major sphincteric abnormality (bladder exstrophy, epispadias), ectopic ureteric orifice	Rare ; men sometimes call severe SUI, UII or MUI "continuous"
With chronic retention	Retention with failure to empty; significant residual ≥ 300 mL	"Overflow incontinence" is no longer an ICS-recommended term . Dangerous in spinal cord injury (SCI) at or above T6 — risk of autonomic dysreflexia

- **Nocturnal enuresis vs nocturia** — enuresis is **wetting while asleep and unaware**; nocturia is **waking to pass urine** during the main sleep period.
- **Other types: postural** (with change of body position); **insensible** (unaware how, but aware of being wet); **postmicturition leakage/dribble** (after micturition, once dressed and usually after leaving the toilet); **functional** (cannot reach the toilet because of cognitive, functional or mobility problems, with a normal lower urinary tract — elderly or disabled); **situational** — coital and giggle incontinence, described in females, rare in males.

TABLE 106.1 CLASSIFICATION OF LOWER URINARY TRACT SYMPTOMS

Storage	Voiding	Postmicturition
Urgency	Slow stream	Postmicturition dribbling
Urinary incontinence	Splitting/spraying	Feeling of incomplete emptying
Increased daytime frequency	Intermittency	
Nocturia	Hesitancy	
	Straining	
	Terminal dribbling	

- **Purpose:** places incontinence among the **storage** symptoms.
- **Trap:** **terminal dribbling** is a **voiding** symptom, but **postmicturition dribbling** and **incomplete emptying** are postmicturition symptoms.

Concept — *Each type is a pressure contest between bladder and outlet: in SUI abdominal pressure overcomes a weak sphincter; in UUI detrusor pressure overcomes a normal outlet; in chronic retention the bladder never empties and pressure stays high — so new bedwetting in an adult man is a retention symptom until proven otherwise.*

EVALUATION

History

- **When** — on coughing, with urgency, asleep, or otherwise. **How often** — nightly, daily, other.
- **Precipitants** — cold weather, **latch-key incontinence** (key in the door).
- **Previous surgery** — prostate, bladder, or major abdominal surgery that may have **damaged the sacral plexus**.
- **How much** — wets underwear, soaks outer clothing or floor. **Containment** — pad type, number, size; condom catheter; spare underwear.
- **Medications tried; neurological symptoms or back pain; sexual, erectile and bowel function; most bothersome symptom.**
- General medical, surgical and social history: drug allergies, smoking, **quantified alcohol and caffeine** intake.

Physical Examination

Sequence (lying): abdomen — scars, masses, **distended bladder**, hernias → external genitalia — foreskin and meatus (**meatal stenosis or severe phimosis can cause UI with chronic retention**) → digital rectal examination (DRE) — prostate size, consistency, tenderness (prostatitis); **pelvic squeeze** to gauge pelvic-floor strength → perineal skin, anal canal, lower rectum — **anal tone and sensation**, rectal emptying → lower-limb neurology — reflexes, spasticity, power, gait, mobility, sensation → mental orientation → **cough test supine, then standing** to detect SUI.

First-Line Investigations

Bedside tests, performed after clinical examination.

Measurement of Height and Weight to Calculate Body Mass Index

- **Body mass index (BMI)** measures obesity; **weight-loss advice may help SUI**, but male evidence is weak — extrapolated from females.

- **With an artificial urinary sphincter (AUS):** the pressure-regulating balloon is usually **61–70 cm H₂O**; intra-abdominal pressure exceeding this is transmitted to the bladder and **the man leaks through the inflated cuff** — less abdominal pressure, less leakage.

Urinalysis

- **Dipstick:** leucocytes and nitrites (infection), glucose (diabetes), blood (haematuria), specific gravity (hydration), pH, ketones.
- **Inexpensive, vital preliminary test in all guidelines** despite false results. Abnormal → **microscopy, culture and sensitivity**; haematuria → cytology, cystoscopy, upper-tract imaging per guidelines.

Bladder Diary

- **Objective record** of leaks, pads, voiding pattern, day and night frequency (giving **mean and maximum voided volume** and functional capacity), fluid type and volume, **nocturnal polyuria**, urgency.
- **Only fully validated diary: International Consultation on Incontinence Questionnaire Bladder Diary (ICIQ-BD).**
- **3 days gives the same information as 7** without exhausting the patient (confirmed by ICIQ-BD validation); include **weekdays and a weekend**, representative of normal life.

Fig. 106.1 ICIQ-BD. Three day-sheets in hourly rows; columns for time, drinks (amount and type), urine output (mL), bladder sensation and pads. "BED" and "WOKE" are written in the time column; a leak is recorded as "LEAK". **Bladder sensation codes:** 0 — no need, voided for "social reasons"; 1 — normal desire, no urgency; 2 — urgency, passed before reaching the toilet; 3 — urgency, reached the toilet without leaking; 4 — urgency, leaked.

Quality-of-Life Questionnaires and Patient-Reported Outcome Measures

- UI causes **little morbidity but major QoL impact** → assess with a **validated questionnaire**.
- **International Prostate Symptom Score (IPSS; American Urological Association Symptom Index, AUA-SI)** is the commonest male questionnaire but **does not measure incontinence**.
- **ICIQ Urinary Incontinence Short Form (ICIQ-UI-SF)** — simple, short, **differentiates SUI from UUI**.
- **ICIQ Male LUTS (ICIQ-MLUTS)** — long form (Fig. 106.3, online only); covers the storage and voiding items of the IPSS and the **bother of each**.
- Diary and questionnaire **repeated after treatment measure its effect**.

Fig. 106.2 ICIQ-UI-SF. Refers to the past four weeks. Item 3 frequency (0–5), item 4 amount (0, 2, 4, 6), item 5 interference with daily life (0–10); **ICIQ score = items 3 + 4 + 5 (0–21)**. Item 6 is unscored and localises type: before reaching the toilet (urgency), on cough or sneeze and when active (stress), when asleep, after dressing (postmicturition), no obvious reason, all the time (continuous).

Pad Testing

- **Weigh an absorbent pad before and after** a specified period under standardised conditions.
- **Unresolved:** 1 or 24 hours, how many days, predictive value. **Does not distinguish UUI from SUI**.
- **Most guidelines do not recommend it;** useful to **quantify leakage** and plan treatment.

GRADING SEVERITY — PAD COUNT VS PAD WEIGHT

Severity	Pads per day	Pad weight gain per day
Mild	1	<200 g
Moderate	2–4	200–400 g

Severity	Pads per day	Pad weight gain per day
Severe	>4	>400 g

- **Pad weight is the better measure** — men change pads for hygiene rather than necessity. **Neither set of cut-offs is properly validated.**

Urine Flow Rate and Postvoid Residual

- Flow rate plus ultrasound **postvoid residual (PVR)**: noninvasive, inexpensive, easy — **interpretation is the skill.**
- **Fast-rising curve** → OAB (urgency) vs **slow rise with long falling tail** → **bladder outflow obstruction (BOO)** from an enlarged prostate.
- **High PVR** → **chronic retention**; with high filling pressures → **nocturnal enuresis.**
- **Flow alone gives no detrusor pressure** → cannot separate BOO from detrusor underactivity.
- **At least two flows** traditionally; the ICS Good Urodynamic Practice document (2017) accepts **one flow if the man confirms it was representative** — so ask him.

Measurement of Prostate-Specific Antigen

- **Prostate-specific antigen (PSA)** is recommended in men with LUTS and **life expectancy >10 years** in whom a cancer diagnosis **would change management**; unnecessary if it would not and the DRE is normal.

Blood Tests

- **Tailored: high PVR or diabetes** → **renal function and glucose.** Index UII or SUI with low PVR → not indicated.
- Clinicians may still test, since **renal insufficiency is otherwise hard to recognise.**

Advanced Investigations

- **Imaging** (plain radiography, ultrasound) — **not routine**; only for suspected upper-tract abnormality, high PVR or urolithiasis.
- **Cystourethroscopy** — suspected **urethral stricture, BOO, blood on urinalysis, or high PVR.**
- **Urodynamic studies (UDS)** — **gold standard** for lower urinary tract (LUT) function. Strictly this includes flow rate; in practice it means **filling cystometry and pressure-flow studies**, with quality control, analysis and documentation.
- **Governing principle: perform UDS only if the result could alter management.** International Consultation on Incontinence (ICI), European Association of Urology (EAU) and American Urological Association (AUA) all advise **conservative and medical therapy first, UDS before invasive therapy.**
- **Other indications:** identify contributing factors; predict upper-tract consequences and intervention outcomes; understand effects of intervention; investigate treatment failure.

Filling Cystometry

- **Bladder** (index non-neurogenic patient): normal, DO, or poor compliance. Leak with a DO wave = **detrusor overactivity incontinence (DOI)**; mark urgency and provocative manoeuvres on the trace.
- **Urethra (EUS)**: competent (continent) or incompetent. **Valsalva or repeated coughs, upright, at 200 mL**; if no leak, repeat **at maximum capacity.** Leak → record the **abdominal (Valsalva or cough) leak-point pressure.**

STRESS INCONTINENCE VS URODYNAMIC STRESS INCONTINENCE VS COUGH-ASSOCIATED DOI

Term	Meaning
SUI	A symptom — clinical diagnosis before UDS
Urodynamic stress incontinence (USI)	A urodynamic observation — leakage on raised intra-abdominal pressure without detrusor contraction
Cough-associated DOI	Cough triggers an involuntary detrusor contraction and the leak is from DO — sounds like SUI clinically

Resolving USI vs cough-associated DOI: fill **supine** (DO less likely) → cough → leak without DO = **USI confirmed**; no leak = **DOI alone**; supine leak without DO in a man who also has DOI = **both (mixed, urodynamically proven)** → invasive treatment offered once the diagnosis is confirmed.

KEY POINTS: INITIAL EVALUATION OF PATIENTS WITH URINARY INCONTINENCE

- **Essential:** history and examination; height and weight for BMI; dipstick urinalysis; bladder diary (e.g., ICIQ-BD); QoL questionnaire (e.g., ICIQ-UI-SF); flow rate; PVR.
- **Optional:** PSA; blood tests (e.g., creatinine); upper-tract imaging; endoscopy; **invasive urodynamics**.

Concept — *Evaluation* escalates from cheap, noninvasive tests that every man receives to invasive urodynamics justified only when its answer would change treatment — *in practice, immediately before surgery*.

PRINCIPAL OPTIONS FOR TREATING MALES WITH URINARY INCONTINENCE

- **Goals:** control continence and **improve QoL**. Three tiers: conservative, pharmacological, surgical.

TREATMENT TIERS

Tier	Options and key points
Conservative	Lifestyle: healthy eating; weight loss in the overweight, obese and diabetic; smoking cessation; less alcohol. Bladder training: scheduled voiding ± double voiding — reduces UI, improves capacity; possibly equivalent to drug therapy. Pelvic floor muscle training (PFMT): effective first line, mainly in post-prostatectomy incontinence (PPI); better if started preoperatively; more effective combined with biofeedback and electrical stimulation
Pharmacological	Antimuscarinics; β_3 -adrenergic agonists; serotonin–noradrenaline reuptake inhibitors. Also α_1 -adrenergic antagonists (α -blockers), phosphodiesterase type 5 inhibitors (PDE5Is), 5- α reductase inhibitors (5ARIs)
Minimally invasive	Refractory storage symptoms: intradetrusor onabotulinumtoxinA , posterior tibial nerve stimulation (PTNS) , sacral neuromodulation (SNM) . SUI: periurethral bulking
Traditional surgery	Male slings, AUS, augmentation enterocystoplasty, continent or incontinent urinary diversion

MANAGEMENT OF URINARY INCONTINENCE IN MALES**Treatment of Urgency Urinary Incontinence****Conservative**

- **Reduce fluid intake by 25%**, provided intake is **>1 L/day**; stop smoking; lose weight; **avoid caffeinated and carbonated drinks**.
- Add **bladder training and PFMT**. Continue **at least 6 weeks, ideally 3 months**.

Antimuscarinics

- **Traditional first-line class** absent contraindications. Previous advice: **at least two agents, ≥4 weeks each**, titrated from low to maximum dose; newer evidence supports **one agent, then moving on** to other management.
- **Class-wide similar efficacy; extended-release preferred to immediate-release.** All have **level 1 evidence, grade A recommendation**. Choice depends on national licensing, local guidelines and preference.
- **Off-label escalation** in selected patients: combining two antimuscarinics or doubling the dose.
- **Anticholinergic burden:** use for **≥3 months increases dementia risk by about half** (systematic review), plus falls and fractures → higher mortality and resource use.

ANTIMUSCARINICS — THE SEVEN MARKETING AGENTS

Agent	Route	Distinguishing feature
Oxybutynin	Oral; topical gel (United States); skin patch	Only transdermal option
Tolterodine	Oral	Usual comparator
Fesoterodine	Oral	Better efficacy than tolterodine but more withdrawal for side effects
Solifenacin	Oral	Better efficacy and less dry mouth than tolterodine
Darifenacin	Oral	—
Propiverine	Oral	—
Trospium chloride	Oral	Nonselective quaternary amine; limited blood–brain barrier penetration → no cognitive effects

β₃-Adrenergic Agonists

- Initially for antimuscarinic intolerance or failure; now **co-first-line**.

MIRABEGRON VS VIBEGRON

Feature	Mirabegron	Vibegron
Generation	First	Second, highly selective
Pharmacology	Lipophilic, hepatic metabolism , >50% excreted in urine	No cytochrome P450 (CYP) interaction → fewer drug–drug interactions
Blood–brain barrier	—	Does not cross → favoured in cognitive impairment
Dose	50 mg (in combination trial)	75 mg once daily (50–100 mg by country)
Licensing	United States, United Kingdom and Europe, Japan	Some countries; Food and Drug Administration (FDA) -approved for adult OAB
Evidence	Superior to tolterodine for incontinence episodes and frequency	Fewer UUI episodes, larger voided volume; favourable side effects vs placebo and anticholinergics; cost-effective to some payers
Signature toxicity	Hypertension → palpitations, headache → monitor blood pressure	Favourable profile

- **Antimuscarinic vs β₃-agonist:** similar efficacy; β₃-agonists cause **less dry mouth and constipation**. With cognitive impairment or dementia risk → **β₃-agonist first line, or early surgical intervention**, rather than an anticholinergic.

- **Combination antimuscarinic + β_3 -agonist** — endorsed by AUA/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction (SUFU) guidelines; different receptors. Phase II: mirabegron + solifenacin **superior to monotherapy**. Phase IIIB: **mirabegron 50 mg + solifenacin 5 mg superior to escalating solifenacin** for efficacy and side effects.
- **Antimuscarinics and β_3 -agonists are the first-line pharmacological treatment for UUI.**
- **Other combinations** (AUA, EAU, National Institute for Health and Care Excellence, NICE): **α -blocker + antimuscarinic** or **α -blocker + β_3 -agonist** for moderate-to-severe **mixed storage and voiding LUTS**.
- **PDE5Is** improve LUTS and UI: relax prostatic, bladder and urethral smooth muscle and improve bladder blood flow. **Tadalafil is the only FDA-approved PDE5I for voiding LUTS in men with erectile dysfunction.**
- **Conservative and medical therapy are usually started together** for quicker relief.

After Failed Conservative and Medical Therapy

- **Invasive UDS** is valuable but **not mandatory before minimally invasive surgery**; if DO or DOI is confirmed, offer minimally invasive treatment where available. Whether UDS should precede onabotulinumtoxinA or SNM is **unproven** and is governed by reimbursement and regulatory bodies.
- **FUTURE trial** (United Kingdom) — UDS before minimally invasive treatment in **women** with refractory OAB; recruitment complete, **results awaited**, likely to be extrapolated to men.

Minimally Invasive Therapy

ONABOTULINUMTOXINA VS SACRAL NEUROMODULATION VS POSTERIOR TIBIAL NERVE STIMULATION

Feature	OnabotulinumtoxinA	SNM	PTNS
Delivery	Cystoscopic intradetrusor injection	Lead in S3 foramen , implanted generator	Needle behind the medial malleolus on the tibial nerve → S2–S4 plexus → sacral micturition centre
Schedule	Repeat every 6–9 months	Two-stage implant; battery replacement or recharging	30 min weekly × 12 weeks , then every other week or monthly
Efficacy	100 U gives greater relief at 12 weeks than most licensed oral doses (network meta-analysis)	Most effective and durable ; cure in about a third, ≥50% improvement in about two thirds; durable beyond 10 years	Equal to antimuscarinics with fewer side effects; better for dry than wet OAB
Key drawback	Urinary retention (about 10% of men) — must accept intermittent self-catheterisation	Most expensive long term	Frequent office visits; no agreed maintenance protocol ; no long-term randomised data
Invasiveness	Intermediate	Most invasive	Least invasive , outpatient

- **No head-to-head trials**; the comparison rests on two meta-analyses.
- **OnabotulinumtoxinA is the only licensed botulinum toxin A formulation: 100 U for idiopathic OAB-wet, 200 U for neurogenic DO.**

BOTULINUM TOXIN A FORMULATIONS

Formulation	FDA status (urology)	Notes
OnabotulinumtoxinA (Botox)	Licensed	Reference formulation
AbobotulinumtoxinA (Dysport)	Not approved	Conversion 2.5:1 to onabotulinumtoxinA; OAB and neurogenic bladder; similar efficacy, more retention
IncobotulinumtoxinA (Xeomin)	Not approved	Neurogenic bladder in some countries; good efficacy

SNM technique: test phase — percutaneous **temporary lead** + external stimulator for **1–2 weeks**, placed under fluoroscopy in the **S3 foramen** → success = **≥50% symptom improvement** → **permanent generator** implanted as second stage. Alternative first stage: **permanent tined lead** + temporary external generator → longer test (**2–4 weeks**), but removal after a failed trial needs a second operation.

- **Mechanism unclear** — thought to modulate the bladder's nerve supply.

SNM GENERATORS

Generator	Battery life	Notes
Earlier generations	5–7 years (use-dependent)	Nonrechargeable
Newer nonrechargeable (2022)	≥15 years	—
InterStim Micro (Medtronic)	About 15 years, weekly recharge	Small, MRI-compatible, rechargeable
Axonics System	About 15 years	First FDA-approved MRI-compatible rechargeable SNM ; most improved at 2 years, over a quarter completely dry

- **PTNS selection** (AUA/SUFU): consider in **drug-naïve patients who decline pharmacotherapy**; recommended for **moderately severe baseline incontinence and frequency with willingness to attend frequent visits**. Home self-administration is possible in theory.
- **PTNS evidence:** safe and effective for OAB with fewer incontinence episodes (meta-analysis); a later systematic review found improvement in frequency, maximum voided volume, urgency and nocturia but **not incontinence** — i.e., **dry OAB rather than wet**; mainly female data.
- **Newer tibial devices:** implantable PTNS (no OAB data yet); **transcutaneous, noninvasive** stimulator (positive early results).

Refractory UUI

- **Containment:** pads, long-term suprapubic or urethral catheter.
- **Major surgery** for the significantly bothered: **augmentation enterocystoplasty** or **ileal conduit ± subtotal cystectomy**. Poor-quality evidence, but **protects the upper tracts and improves QoL**; **high short- and long-term morbidity**.
- **Autoaugmentation is no longer recommended** for adult DO — poor long-term success and complications.

Concept — *UUI is managed as a ladder of increasing invasiveness — lifestyle, drugs, neuromodulation or toxin, reconstruction — where each rung is climbed only on failure of the last, and cognitive safety now shapes the drug rung as much as efficacy does.*

Treatment of Stress Urinary Incontinence

Sequence: supervised **PFMT for ≥3 months** (pelvic-floor physiotherapist) → ± duloxetine → rule out urethral or anastomotic stricture and bladder dysfunction → surgery.

- **Commonest cause of male SUI: prostatectomy.** After radical prostatectomy, **no surgery for at least 6–12 months** — continence continues to improve spontaneously.
- **While waiting:** penile clamp (Dribble-Stop, Pacey cuff); or, more often, condom catheter, pads, suprapubic catheter, or urethral catheter — **not long term (traumatic hypospadias)**. Catheter routes do not differ in urinary tract infection (UTI) or upper-tract change, but **suprapubic causes fewer urethral complications**.
- **Duloxetine** (serotonin–noradrenaline reuptake inhibitor): **off-label in men** — licensed in some countries only for women with moderate-to-severe incontinence; male data limited and heterogeneous; **short-term success in most post-prostatectomy men** despite discontinuation for side effects; long-term efficacy unknown.

- **AUS = gold standard for moderate-to-severe SUI, about 80% continence**; first described by **Scott in 1973**.
- **Mild-to-moderate alternatives** (variable efficacy): urethral bulking agents, male slings (fixed and adjustable), adjustable balloon continence therapy (ProACT).
- **Bulking agents** enhance sphincter coaptation: **inferior, with no proven long-term efficacy whatever the material**; short-term improvement with transient haematuria and dysuria. Reserve for minimal leakage or men unfit for more invasive surgery; **guidelines advise against them for PPI**.

TABLE 106.2 SECOND- AND THIRD-LINE THERAPY RECOMMENDATIONS FOR MALES WITH STRESS URINARY INCONTINENCE

Intervention	Recommendation	Grade
Duloxetine	Offer to males with SUI	Weak
	Inform of possible adverse events and off-label use	Strong
Bulking agents	Do not offer for PPI	Weak
Fixed sling	Offer nonadjustable transobturator sling for mild-to-moderate PPI	Weak
	Inform that severe incontinence, prior pelvic radiotherapy or transurethral surgery may worsen outcome	Weak
AUS	Offer for moderate-to-severe PPI	Strong
AUS or ProACT	Implant only in expert centres	Weak
	Warn of high risk of complications, mechanical failure or explantation , even in expert centres	Strong
ProACT	Do not offer this noncircumferential device after pelvic radiotherapy	Weak

Data from EAU guidelines 2023: Management of non-neurogenic male lower urinary tract symptoms (LUTS).

- **Purpose:** maps the post-PFMT options by severity. **Pattern:** only three Strong grades — inform about duloxetine, **offer AUS for moderate-to-severe PPI**, warn about AUS/ProACT complications; all else Weak.
- **Traps:** bulking is a "do not offer"; ProACT is **contraindicated after radiotherapy**; radiotherapy, severe leakage and prior transurethral surgery **degrade fixed-sling results**.

Male Slings

- **Good efficacy, acceptable morbidity in mild-to-moderate SUI**; transobturator or retropubic.
- **AdVance** (fixed): polypropylene mesh, **transobturator, under the membranous urethra**; cure rates vary widely because **definitions of success differ**.
- **Adjustable slings** allow postoperative tension adjustment; three main systems.

ADJUSTABLE MALE SLINGS

System	Design	Outcome pattern
Remeex	Adjustable sling; readjusted in about a third	High objective success regardless of severity ; low complication and removal rates at medium term
Argus-T	Adjustable sling	Success highest in mild, lower in severe SUI ; readjustment common; success falls by 5 years
Adjustable transobturator male system (ATOMS)	Inflatable silicone cushion compressing the urethra; titanium port (lower abdomen or scrotum) — a noncircumferential sphincter	High objective cure with 2–4 adjustments ; explantation about one in five; third generation (preattached silicone-covered scrotal port) — better durability, fewer adjustments

- **Adjustable vs fixed:** adjustable may give higher objective cure but **more complications and explantation**; heterogeneous data → **both acceptable, neither superior**.
- **MASTER trial** (Men with Urodynamic Stress Incontinence After Prostate Surgery; noninferiority randomised controlled trial, RCT) — **AUS vs male sling**: both improved symptoms, satisfaction and QoL;

equivalent at 12 months; secondary analysis favoured **AUS** (better QoL, fewer pads and leaks, higher satisfaction); under a strict definition **only 15% in either arm never leaked**.

- **ProACT: two compressive balloons at the bladder neck**, adjusted via scrotal ports; continence in up to about two thirds, **explantation in up to about half**. **ATOMS is more efficacious with higher satisfaction than ProACT**.
- **Neurogenic SUI** often needs **more than one intervention** — coexistent low compliance and neurogenic DO.

Concept — *Male SUI is a sphincter injury that often recovers, so the rule is wait and train first, then match the device to severity and to radiotherapy status: slings and balloons for lesser leakage in unirradiated tissue, the AUS for severe leakage.*

Treatment of Mixed Urinary Incontinence

- **Treat the most bothersome component first**; more challenging than either pure type.
- The SUI component may be **hidden, even on UDS, when UUI predominates** → treat DO first (drugs or minimally invasive) → **repeat UDS** to reveal any SUI.

Treatment of Other Types of Urinary Incontinence

- **Treat the underlying cause.**
- **High-pressure, low-flow retention with nocturnal enuresis** → **catheterise first** to decompress → then treat BOO: endoscopic resection, vaporisation or enucleation, or open or robotic prostatectomy.
- **Nocturnal enuresis without residual** → usually OAB → antimuscarinic or β_3 -agonist, possibly **desmopressin**. After cystoprostatectomy with **neobladder**, night-only leakage from pelvic-floor relaxation in sleep → conservative.
- **Postmicturition dribble**: poorly studied, **no approved drug**; treat with **PFMT (strong squeeze at end of voiding) plus urethral milking**.

Urethral milking: wait a few seconds after voiding → fingertips **three fingerbreadths behind the scrotum**, gentle midline pressure → move forward under the scrotum to the penile base, pushing urine into the penile urethra → milk, squeeze and shake the penis → **repeat twice**.

ALGORITHMS FOR MANAGEMENT OF MALES WITH URINARY INCONTINENCE

Fig. 106.4 International Consultation on Incontinence Algorithm

From Cardozo L, Rovner E, Wagg A, et al.: *Incontinence*, 7th ed, 2023, ICUD; 2159–2161.

FIG. 106.4 (A) INITIAL MANAGEMENT OF URINARY INCONTINENCE IN MALES

Presentation	Presumed diagnosis	Initial treatment (level of recommendation)
Postmicturition dribble	—	Urethral stripping or milking (B) ; PFM contraction (C)
Incontinence on exertion (usually post-prostatectomy)	SUI — sphincteric incompetence	Discuss options: lifestyle; PFMT ± biofeedback (B) ; transcutaneous or percutaneous tibial nerve stimulation (B); scheduled voiding/bladder training in OAB (C);
Mixed symptoms	MUI — treat most bothersome first	antimuscarinic/β_3-agonist for OAB ± UUI (B); α-blocker ± 5ARI if BOO suspected
Urgency/frequency ± UUI	UUI — DO	

Presentation	Presumed diagnosis	Initial treatment (level of recommendation)
"Complicated": recurrent or "total" incontinence; with pain, haematuria, recurrent infection, prostate irradiation, radical pelvic surgery; or any abnormality (e.g., significant PVR)	—	Straight to specialised management
• Common assessment mirrors the evaluation above (diary, questionnaire, QoL, examination, urinalysis ± culture — treat infection and reassess — pelvic-floor function, PVR). Failure → specialised management; containment products throughout.		

FIG. 106.4 (B) SPECIALISED MANAGEMENT OF URINARY INCONTINENCE IN MALES

Diagnosis (after UDS, imaging, urethrocystoscopy if indicated)	Treatment (level of recommendation)
SUI (sphincteric incompetence; post-prostatectomy)	If initial therapy fails: AUS (A) — grade applies only to the AMS 800 , as newer devices lack its evidence base — or male sling (C)
MUI	Treat the major component first
UUI — DO during filling	If initial therapy fails: botulinum toxin A ; sacral nerve stimulation (B)
UUI with coexisting BOO	α -blocker, 5ARI (C); correct anatomical BOO (C) ; antimuscarinic/ β_3 -agonist (B)
UUI with underactive detrusor during voiding	Intermittent catheterisation + antimuscarinic/ β_3 -agonist
"Complicated" (recurrent/total; prostate or pelvic irradiation; radical pelvic surgery)	Urethrocystoscopy, further imaging, UDS → correct anomaly, treat pathology

Fig. 106.5 EAU Treatment Algorithm of Males With Urinary Incontinence

From Burkhard FC (Chair), Bosch JLHR, Cruz F, et al.: EAU Pocket Guidelines on Urinary Incontinence, 2020, pp 192–193.

FIG. 106.5 EAU ALGORITHM — STEPS AND STRENGTH OF RECOMMENDATION

Step	Content (strength)
Initial assessment	History, examination, urinalysis (Strong); questionnaire and voiding diary when standardised assessment is required (Strong); PVR if voiding difficulty (Strong); pad test (Weak)
Specialist referral triggers	Haematuria, pain, recurrent UTI, previous pelvic radiotherapy, abnormal DRE, suspected voiding dysfunction
Initial management	Individualised behavioural and physical therapy including PFMT; advice on bowel function, drugs, comorbidity, fluid intake; weight loss; pads or containment; timed or prompted voiding in elderly or care-dependent people (all Strong). Urgency or mixed: antimuscarinic or mirabegron (Strong)
After failure	Consider urodynamics and LUT assessment (Strong)
Urgency / urgency-predominant mixed	OnabotulinumtoxinA or sacral nerve stimulation (Strong — evidence mainly female) ; bladder augmentation or urinary diversion (Weak)

Step	Content (strength)
Stress / stress-predominant mixed	Fixed sling or AUS for PPI (Weak)

ICI VS EAU ALGORITHMS — THE SAME DECISIONS COMPARED

Decision	ICI (Fig. 106.4)	EAU (Fig. 106.5)
Entry split	By presentation ; separate dribble and "complicated" streams	By red flags → specialist; then by type
Pad test	Not listed	Weak
Drugs	Antimuscarinic/ β_3 -agonist (B); α-blocker $\pm$ 5ARI if BOO	Antimuscarinic or mirabegron (Strong)
Tibial nerve stimulation	In initial management (B)	Not listed
Refractory UUI	Botulinum toxin A, sacral nerve stimulation (B); stratified by BOO and detrusor underactivity	OnabotulinumtoxinA or SNS (Strong*); augmentation/diversion (Weak)
SUI surgery	AUS (A) , sling (C)	Fixed sling or AUS (Weak)

- **Trap:** AUS strength differs by document — **ICI grade A; EAU 2020 pocket algorithm Weak** (sling or AUS); **EAU 2023 Table 106.2 Strong** for moderate-to-severe PPI.

Concept — *Both algorithms share one spine* — **simple assessment, empirical conservative and drug therapy by presumed type, then urodynamics and specialist intervention on failure, with red-flag or "complicated" cases diverted straight to specialist care.**

CONCLUSIONS

- **Assess to identify the cause**, then treat **conservatively and medically first**; **reserve surgery** for failure of initial treatment when UI significantly affects QoL.

KEY POINTS: TREATMENT OF URINARY INCONTINENCE

- **UUI:** fluid manipulation and lifestyle; bladder training and PFMT; antimuscarinics and/or β_3 -agonists; intradetrusor onabotulinumtoxinA; SNM; PTNS; augmentation enterocystoplasty; urinary diversion (e.g., ileal conduit).
- **SUI:** weight loss and lifestyle; PFMT; duloxetine (off-licence); penile clamp; **AUS (gold standard)**; suburethral sling (fixed or adjustable); ProACT balloon.